

Name: K Vamsi

Reg no:192372062

TITLE:HistoQuanta: Intelligent Oncology Assistant for Diagnosis and Treatment Planning

Description :

HistoQuanta is an advanced oncology-focused mobile application designed to assist doctors and cancer specialists in performing accurate and efficient cancer-related calculations. The platform integrates automated calculation tools, secure patient data management, and real-time analytics to support clinical decision-making.

Traditionally, oncologists rely on manual calculations, spreadsheets, and reference materials for determining chemotherapy dosages, cancer staging, and treatment planning. These methods are time-consuming and prone to human error. HistoQuanta addresses these challenges by providing a centralized digital solution that enhances precision, reduces workload, and improves treatment outcomes.

The system consists of an Android-based application developed using Kotlin, supported by a backend built with PHP, XAMPP, and MySQL for secure data storage and processing. It includes an interactive calculation engine for dosage computation and tumor analysis, along with an analytics module that helps predict treatment effectiveness and monitor patient progress.

With its intuitive UI/UX design (developed using Figma), HistoQuanta ensures ease of use for medical professionals. The application enables doctors to access patient data, perform calculations, and make informed decisions anytime and anywhere.

Overall, HistoQuanta provides a modern, reliable, and efficient solution that bridges the gap between traditional oncology practices and technology-driven healthcare systems, ultimately improving accuracy, saving time, and enhancing patient care.



